BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

In the Matter of

The Brooklyn Union Gas Company d/b/a National Grid NY &
KeySpan Gas East Corporation d/b/a National Grid

Case 19-G-0309 & 19-G-0310

August 2019

Prepared Testimony of:

Staff Efficiency and
Sustainability Panel

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Q. Members of the Staff Efficiency and Sustainability Panel (Panel), please state your names, employer and business address.

A. Our names are Robert Cully, Joseph Hitt, Laurie Kokkinides, Kathryn Mammen, Davide Maioriello, and Jennifer Park. We are employed by the New York State Department of Public Service (Department) and our business address is Three Empire State Plaza, Albany, New York 12223.

Q. Mr. Cully, what is your position at the Department?

A. I am currently employed as a Utility Engineering Specialist 3 assigned to the Energy Efficiency and Demand Response Section of the Office of Markets and Innovation (OMI).

Q. Mr. Cully, describe your educational background.

A. I graduated from Clarkson University with a Bachelor of Science degree in Environmental Engineering in May 2010.

Q. Mr. Cully, describe your professional experience and your position at the Department.

A. I have been employed by the Department since August 2012. Since joining the Department, my responsibilities have included examining utility

rates and fees, rate design, benefit cost analyses, demand response programs, electric standby service rates, demand management programs, including Non-Wires Alternatives, and Earning Adjustment Mechanisms (EAMs).

Q. Mr. Cully, have you previously provided testimony before the Public Service Commission (Commission)?

A. Yes. I have provided testimony in numerous rate proceedings. Most recently, I provided testimony in Cases 13-E-0030, 13-G-0031, and 13-S-0032, 16-E-0060, 19-E-0065, and 19-G-0066, Consolidated Edison Company of New York, Inc. (Con Edison); Cases 14-E-0493, 18-E-0067, and 18-G-0068, Orange and Rockland Utilities, Inc. (Orange and Rockland); Case 17-E-0459 and 17-G-0460, Central Hudson Gas & Electric Corporation (Central Hudson); Cases 17-E-0238 and 17-G-0239, Niagara Mohawk Power Corporation d/b/a National Grid (Niagara Mohawk); and Cases 15-E-0283 and 15-E-0285, New York State Electric & Gas Corporation and Rochester Gas and Electric Corporation.

Q. Mr. Hitt, what is your position at the

Department?

A. I am employed as a Utility Analyst II assigned to the Clean Energy section of the Office of Markets and Innovation.

Q. Please provide a summary of your educational background and professional experience.

A. I graduated from the State University of New York Institute of Technology at Utica/Rome in 1998 with a Bachelor of Science degree in Industrial Engineering Technology, where I also received a Master of Science degree in Advanced Technology in 2000. Additionally, I received a Master of Business Administration degree from the University at Albany in 2006. Prior to and while pursuing my education, I held several technical positions of increasing responsibilities. My technical skills and education led to Operations Management positions in the transportation and energy industries. I joined the Department in July 2012. My responsibilities at the Department include reviewing and monitoring utility energy efficiency programs, the Clean Energy Fund, and a monitoring and auditing role of the energy

efficiency programs Technical Resource Manual.

Q. Have you previously provided testimony in proceedings before the Commission?

A. Yes, I provided testimony in the following rate cases: Case 16-W-0259, New York American Water Company, Inc.; and Cases 16-E-0061 and 16-G-0061, Con Edison.

Q. Ms. Kokkinides, what is your position in the Department?

A. I am employed as a Utility Analyst 1 assigned to the Energy Efficiency section of OMI.

Q. Please describe your educational background.

A. I received a Bachelor of Arts degree in Environmental Studies and in Geography from Binghamton University in 1994 and a Master of Arts degree in Geography from Binghamton University in 1996.

Q. Ms. Kokkinides describe your professional experience and responsibilities with the Department.

A. I have been employed by the Department since 2018. Prior to joining the Department, I held positions in private industry and at the New York State Energy Research and Development

Authority that involved site research and investigations, development and evaluation of energy efficiency programs and development of policy implementation metrics and impacts. Currently, my relevant responsibilities include reviewing and monitoring utility energy efficiency programs, evaluation, monitoring and verification (EM&V) policy review and Clean Energy Guidance.

Q. Have you previously testified before the Public Service Commission (Commission)?

A. No. I have not testified before the Commission.

Q. Ms. Mammen, what is your position at the Department?

A. I am a Utility Analyst 3 in the Clean Energy Section of OMI.

Q. Please describe your educational background.

A I received a Bachelor of Arts Degree in sociology from Fordham University in 2006. I also received a Master of Public Administration from Rockefeller College of Public Affairs & Policy at the University at Albany, State University of New York in 2010.

Q. Please describe your professional experience and

responsibilities at the Department.

A. I joined the Department in 2009, first as an intern for the Secretary to the Commission, and later as a Utility Analyst in the Office of Energy Efficiency and the Environment where I was responsible for overseeing the implementation and evaluation of energy efficiency programs offered under the Energy Efficiency Portfolio Standard. Currently, my responsibilities include reviewing and monitoring utility energy efficiency programs and developing policy recommendations for Commission consideration related to energy efficiency and other clean energy activities.

Q. Have you previously provided testimony before the Commission?

A Yes. Most recently, I testified in Cases 19-E-0065 and 19-G-0066, Con Edison. I have also testified in Cases 15-G-0058 and 15-G-0059, KeySpan Gas East Corporation d/b/a National Grid (KEDLI) and The Brooklyn Union Gas Company d/b/a National Grid NY (KEDNY) on energy efficiency issues; and in Case 16-W-0130, Suez Water New York Inc., on the issue of the water

conservation program.

Q. Mr. Maioriello and Ms. Park, do you discuss your credentials in another testimony submitted in these proceedings?

A. Yes. We discuss our credentials in the direct testimony of the Staff Gas Supply Panel.

Q. Panel, what is the purpose of your testimony in these proceedings?

A. We will address KEDNY's and KEDLI's (jointly, the Companies) proposals and our recommendations regarding: (1) gas energy efficiency portfolios, including increased budgets and savings targets for the Company-administered gas portfolios, cost recovery mechanisms necessary to fund such portfolios, the KEDLI HEAT Program, and the Companies' proposed additional full-time employees (FTEs) to manage the Companies' expanded clean energy activities; (2) the Utility Energy Services Contract Program; (3)the Fuel Switching Calculator; (4) Demand Response; (5) the Expanded Geothermal Demonstration Projects; (6) the Clean Conversion Program; (7) the Low- and Moderate-Income Gas Conversion Programs; (8) renewable natural gas

interconnections; (9) the Power-to-Gas Demonstration Project; and (10) the Hydrogen Blending Study.

Q. In this testimony, will the Panel refer to, or otherwise rely upon, any information obtained during the discovery phase of these proceedings?

A. Yes. We will refer to, and have relied upon, several responses to Department Staff Information Requests (IRs), all of which are included in Exhibit__(SESP-1). We will refer to these IRs by the designation given when issued, e.g., DPS-123.

Q. Is the Panel sponsoring any other exhibits?

A. Yes. The Panel is sponsoring one additional exhibits in our testimony.

Q. Please describe these exhibits.

A. Exhibit__(SESP-2) shows our recommended use of unspent energy efficiency funds.

Q. What time period does the Panel’s testimony address?

A. Although the Company’s testimony includes proposals for the Rate Year, the 12-month period ending March 31, 2021; Data Year 1, the 12-month period ending March 31, 2022; Data Year 2, the

12-month period ending March 31, 2023; and Data year 3, the 12-month period ending March 31, 2024, our testimony focuses on the Rate Year.

ENERGY EFFICIENCY

Background

Q. Provide a brief background of utility-administered energy efficiency programs beginning with Energy Efficiency Portfolio Standard (EEPS).

A. On June 23, 2008, the Commission issued the Order Establishing Energy Efficiency Portfolio Standard and Approving Programs in Case 07-M-0548. Within the Commission's EEPS framework, New York State's investor-owned electric and gas utilities and the New York State Energy Research and Development Authority (NYSERDA) implemented residential, multifamily, and commercial and industrial energy efficiency programs. The Commission approved each EEPS program, and the associated annual budgets and savings targets, through a series of orders issued in Cases 07-M-0548, 09-G-0363, and other related cases. The second and final phase of the Companies' EEPS

programs ended on December 31, 2015. Funding for the EEPS programs was collected through a surcharge mechanism, the System Benefit Charge.

Q. What actions did the Commission take with respect to energy efficiency programs in preparation for the conclusion of EEPS?

A. First, the Commission authorized a new regime for electric efficiency programs in its February 26, 2015 Order Adopting Regulatory Policy Framework and Implementation Plan issued in Case 14-M-0101 (Track One Order). The Track One Order established portfolio-level budgets and savings targets for electric efficiency programs for 2016 and required all affected utilities to file a Budget and Metric Plan and a companion Energy Efficiency Transition Implementation Plan (ETIP) by July 15, 2015, reflecting the authorized 2016 budgets and targets and proposing budgets and targets for 2017 and 2018.

Q. What followed the Track One Order?

A. Following the Track One Order, on June 19, 2015, the Commission issued the Order Authorizing Utility-Administered Gas Energy Efficiency Portfolios for Implementation Beginning January

1, 2016 in Cases 07-M-0548 and 15-M-0252 (June 2015 Gas ETIP Order). The June 2015 Gas ETIP Order authorized 2016 budgets and targets for gas energy efficiency programs and required utilities to incorporate the gas programs into the July 15, 2015 Budget and Metric Plan and ETIP filings.

Q. Through what mechanism were the ETIP programs to be funded?

A. The June 2015 Gas ETIP Order required utilities to file tariff revisions establishing an Energy Efficiency Tracker (EE Tracker) surcharge mechanism to recover the costs associated with the ETIP energy efficiency programs. In its December 18, 2015 Order Approving Tariff Amendments, the Commission approved the Companies' EE Tracker surcharge mechanisms effective January 1, 2016.

Q. When were the energy efficiency portfolios and implementation plans resolved by the Commission?

A. The Commission acted on the utility Budget and Metrics filings to establish portfolio-level electric and gas budgets and savings targets in its January 22, 2016 Order Authorizing Utility-

Administered Energy Efficiency Portfolio Budgets and Targets for 2016-2018 (January 2016 ETIP Order) in Case 15-M-0252.

Q. How did the Commission's ETIP framework change the way in which the Companies administered efficiency programs?

A. In January 2016, the Companies began implementing gas energy efficiency programs under the Commission's new framework. This new framework provided KEDNY and KEDLI more flexibility in designing and implementing their respective electric and gas efficiency programs to meet portfolio-level Dekatherm (Dth) savings targets within Commission-authorized portfolio-level budgets, as opposed to specific program authorizations.

Q. Has the Commission authorized budgets and targets beyond the 2016-2018 period?

A. Yes, the Commission authorized 2019 and 2020 gas budgets and targets in its March 15, 2018 Order Authorizing Utility-Administered Energy Efficiency Portfolio Budgets and Targets for 2019-2020, issued in Case 15-M-0252 (March 2018 ETIP Order).

Q. Did the March 2018 ETIP Order include other directives associated with the implementation of ETIP programs?

A. Yes. In the March 2018 ETIP Order, the Commission: (1) adjusted each utility's targets from net to gross savings and emphasized the importance of realized savings; (2) changed the units of gas savings targets and reported savings from Dth to Million British Thermal Units (MMBtu); (3) required recovery of costs for utility-administered energy efficiency programs to be shifted into base delivery rates; (4) directed any utility EAM based on energy efficiency programmatic performance be calculated and awarded based on verified gross savings; and, (5) required each utility to evolve its ETIP into a System Energy Efficiency Plan (SEEP).

Q. Please explain the Commission's directive to shift to gross savings targets.

A. Previous Commission policy required that both energy efficiency targets, and savings reported against those targets, be expressed on a net basis, accounting for the effects of free-

ridership and spillover by applying a default net-to-gross ratio of 0.90. In recognition of specific barriers that make it difficult to meaningfully calculate accurate net-to-gross ratios, the Commission determined that, going forward, goals should be established, and savings reported, on a gross basis. The March 2018 ETIP Order defined gross savings as the change in energy consumption and/or demand that results directly from program-related actions taken by participants in a program, regardless of why they participated.

Q. Pursuant to the March 2018 ETIP Order, when were utilities required to shift cost recovery for utility-administered energy efficiency programs into base rates?

A. The Commission directed each utility to move the energy efficiency costs collected through the EE Tracker surcharge mechanism into base delivery rates during the first rate proceeding for that utility following the March 2018 ETIP Order.

Q. Did the Commission provide guidance regarding the reconciliation of energy efficiency costs once those costs are collected through base

rates?

A. Yes. The Commission required a downward-only reconciliation to be conducted at the end of a rate plan, and annually thereafter during any stay-out period, with any unspent funds to be deferred, with appropriate carrying charges, for future ratepayer benefit. Utilities, according to the Commission, are allowed to carry funds from year to year during a rate plan.

Q. Please explain the Commission’s direction regarding verified gross savings.

A. At a high-level, verified gross savings include adjustments associated with evaluation, measurement and verification (EM&V) findings that verify key factors used to calculate savings. The Commission’s requirement for verified gross savings results from its increased focus on actual realized savings, which are critical to expanded energy efficiency and the attainment of the State’s greenhouse gas emission reduction objectives. The direct testimony of the Staff EAM Panel includes additional details and recommendations regarding verified gross savings.

Q. What is the SEEP?

A. The SEEP must describe the entirety of the utility's expanded reliance on, and use of, cost-effective energy efficiency to support its distribution system and customer needs.

Q. Did any Commission actions outside of the Utility ETIP or rate case proceedings occur that affected the Company's energy efficiency portfolios?

A. Yes. The Commission issued its Order Adopting Accelerated Energy Efficiency Targets (December 2018 Efficiency Order) on December 13, 2018, in Case 18-M-0084 (Utility Energy Efficiency Proceeding). In that Order, the Commission increased the Companies' 2019 and 2020 gas targets and budgets and established a process through which the Commission will establish targets and budgets for the 2021-2025 period.

Q. Please describe the Commission's actions taken in the December 2018 Efficiency Order and how they affected KEDNY and KEDLI's energy efficiency portfolios.

A. In that Order, the Commission increased the Companies' 2019 and 2020 gas targets and budgets

and required KEDNY, KEDLI, and the other large investor-owned utilities, in consultation with NYSERDA, to file proposed energy efficiency budgets and targets for the 2021-2025 period on or before March 31, 2019.

Q. What preceded the December 2018 Efficiency Order?

A. In April 2018, Department Staff and NYSERDA issued a report entitled New Efficiency: New York (NE:NY Report). Among other things, the NE:NY Report highlighted the importance of energy efficiency in achieving the State’s goal of a 40% statewide reduction of greenhouse gas emissions from 1990 levels by 2030, and established a 2025 statewide energy efficiency target of 185 trillion Btu (TBtu) of energy usage reductions at the customer level, which required utilities to achieve an additional 31 TBtu reduction above and beyond the then-current energy efficiency program target levels. The NE:NY Report also identified a number of strategies to reach the statewide 185 TBtu savings goal by 2025.

Q. What other relevant actions did the Commission

take in the December 2018 Efficiency Order?

A. In the December 2018 Efficiency Order, the Commission determined the Statewide 185 TBtu target was reasonable and adopted the incremental 31 TBtu as a fuel neutral target for the State's utilities to achieve through their electric and gas efficiency portfolios by 2025. The Commission also established sub-targets of at least 5 TBtu from heat pump adoption. The Commission also established a cap of $1.6 billion to achieve the 31 TBtu; required 20% of incremental spending to fund low- and moderate-income customer programs; and established an expectation for cost reduction per unit of energy saved.

Q. Have the Companies made the filing as required by the December 2018 Efficiency Order?

A. Yes, the Companies' proposals were included in the NY Utilities Report Regarding Energy Efficiency Budgets and Targets, Collaboration, Heat Pump Technology and Low- and Moderate-Income Customers (April Utility Proposal). This report was filed on April 2, 2019, in the Utility Energy Efficiency Proceeding. KEDNY-

and KEDLI-specific proposals were set forth in Chapter XII, the National Grid Chapter. Subsequent to the April Utility Proposal filing, the Utilities have jointly filed revisions to that proposal with the most recent dated May 10, 2019 (May Utility Proposal), which includes the Companies' proposals in a revised Chapter XII and a new appendix, Appendix A (Companies' May Proposal). Our testimony is based upon the contents of the May Utility Proposal, the Companies' May Proposal, and the Companies' Corrections and Updates filing in these rate proceedings filed on July 3, 2019.

Q. Does the Companies' May Proposal align with the Companies' initial testimony provided by the Future of Heat Panel?

A. No. Although the targets provided in both sources matched, there is a discrepancy between the calendar year budgets for 2019 and 2020. However, in the Corrections and Updates Testimony of the Future of Heat Panel, the Companies' corrected the budget amounts the Companies propose to move into base rates. The Companies stated that the corrections to the

budgets were the result of a change in the revenue requirement, however we view these corrections to the budgets as a means of aligning with the December 2018 Efficiency Order and the Companies' May Proposal, not as a function of any changes to the revenue requirement.

Budgets and Targets

Q. What budgets do the Companies propose in their Corrections and Updates filings?

A. On page 7 of the Corrections and Updates testimony of the Future of Heat Panel, the Companies state that they would use the authorized budget for calendar year 2020 and the 2021-2025 presumptive targets and budgets set forth in the December EE Order as the proposed budgets and targets for the Rate Year and Data Years until the Commission directs otherwise. The Future of Heat Panel proposed a Rate Year budget of $18.636 million for KEDNY and $9.478 million for KEDLI.

Q. What are the Companies' targets for calendar years 2019, 2020, 2021, 2022 and 2023?

A. The budgets provided by the Future of Heat Panel on page 22 of its direct testimony, were for the Rate Year, however, the targets provided reflect the following calendar years: 2019-2020, which the Companies combined; 2021; 2022; and 2023. For KEDNY the calendar year 2019-2020 gas savings target is 811,814 MMBtu and for KEDLI it is 452,387 MMBtu. The 2021; 2022; and 2023 MMBtu targets for KEDNY are 510,740; 574,740; and 704,740, respectively. For KEDLI the MMBtu targets for 2021; 2022; and 2023 are 268,821; 301,821; and 343,821, respectively.

Q. Do you agree with the budgets and targets proposed by KEDLI and KEDNY in the Companies' May Proposal and the Future of Heat Panel's Corrections and Updates testimony?

A. We recommend that the Companies' proposed targets and budgets be used as placeholders until the Commission acts on the May Utility Proposal in the Utility Energy Efficiency Proceeding. Following Commission action on that proposal, the final targets and budgets authorized for KEDLI and KEDNY in that proceeding should be adopted in these rate

proceedings. If Commission action in the Utility Energy Efficiency Proceeding does not occur before the Commission sets rates in these proceedings, we recommend establishing a deferral mechanism. This mechanism should reconcile the difference between the revenue requirement impacts approved in these rate proceedings with those ultimately approved in the Utility Energy Efficiency Proceeding. Discussion of the energy efficiency targets as they relate to the Companies' proposed Earnings Adjustment Mechanisms can be found in the Staff EAM Panel's testimony.

Cost Recovery Mechanism

Q. How are the Companies' energy efficiency programs currently funded?

A. The Companies currently fund energy efficiency programs through the EE Tracker, a component of the System Benefit Charge. The EE Tracker is designed to collect the authorized funding for the Companies' current ETIP programs.

Q. How do the Companies propose to treat energy efficiency costs in these rate proceedings?

A. On page 24 of the Future of Heat Panel's direct testimony, the Companies propose to collect energy efficiency funds through base rates.

Q. Does the Panel agree with the Companies' proposal to shift energy efficiency costs out of the EE Tracker and into base rates?

A. Yes. The Companies' proposal to move energy efficiency costs into base rates is consistent with previous Commission direction provided in the March 2018 ETIP Order, which required each utility, during its next rate proceeding, to recover the costs for the portfolio of utility-administered energy efficiency programs through base delivery rates. In addition, shifting the energy efficiency funding out of the EE Tracker and into base rates is consistent with the approach adopted by the Commission in the most recent rate proceedings for Niagara Mohawk, in Cases 17-E-0238 and 17-G-0239; Central Hudson, in Cases 17-E-0459 and 17-G-0460; and Orange and Rockland, in Cases 18-E-0067 and 18-G-0068.

Q. Do the Companies propose any additional changes to the treatment of energy efficiency costs?

A. Yes. On page 24 of the Future of Heat Panel's

direct testimony, the Companies propose several additional changes related to the treatment of evaluation, measurement and verification (EM&V) funds associated with energy efficiency programs.

Q. Explain the Companies proposed changes to the collection of EM&V funds?

A. The Companies propose to: (1) move EM&V costs out of the energy efficiency portfolio budget, (2) maintain the portfolio-level energy efficiency budget, and (3) collect the EM&V funds separately through base rates. The Rate Year EM&V costs that would be shifted out of the energy efficiency budget for KEDNY and KEDLI are $924,353 and $473,898, respectively.

Q. What justifications do the Companies provide for the EM&V funding changes?

A. As explained on pages 24-25 of the Future of Heat Panel’s direct testimony, the Companies’ justification for these changes is that it will enable them to fund additional customer energy efficiency programs. The Companies also state that these costs are either not directly related to delivery of the energy efficiency portfolio

contained in the ETIPs or are scoped to integrate initiatives beyond just energy efficiency.

Q. Does the Panel agree with the Companies' proposals related to the EM&V funding?

A. No. The Companies did not provide enough detail in the need or use of the additional budget. The Companies state that the additional monies will enable them to add additional energy efficiency programs and initiatives that would not increase the ETIP budgets. In the Companies' response to IRs DPS-383 and DPS-384, the Companies state that the latest ETIP, filed February 19, 2019, represents their current implementation plan for their respective portfolios of energy efficiency projects, including those that would be funded with the replaced EM&V costs. The Companies' ETIPs do not identify any additional customer programs. In addition, the Commission is currently considering the Companies' energy efficiency budgets inclusive of EM&V in the Utility Energy Efficiency Proceeding. Therefore, we recommend maintaining EM&V within the Companies' energy

efficiency budgets.

Q. Does the Panel recommend a reconciliation of energy efficiency costs?

A. Yes. We recommend a reconciliation that aligns with the Commission's direction in the March 2018 ETIP Order. Specifically, we recommend a downward-only reconciliation of the energy efficiency expenses at the end of the Rate Year. In the event of a multi-year rate plan, we recommend that the Companies be allowed to carry unspent funds from year to year during the term of the rate plan. At the end of the rate plan, and annually thereafter, any unspent funds should be deferred, with appropriate carrying charges, for future customer benefit.

Q. Do the Companies propose to collect the entirety of the energy efficiency budgets through base rates?

A. No. In the Companies' response to IR DPS-268, the Companies propose to use unspent EEPS funds to offset a portion of the energy efficiency budgets and then collect the remainder of the budgets through base rates.

Q. Please explain what is meant by "unspent EEPS

funds."

A. Unspent EEPS funds are those funds that the Companies collected from customers to pay for energy efficiency programs offered during the previous EEPS period but that the Companies have neither spent nor committed to spend.

Q. How do the Companies propose to use unspent EEPS funds to offset the energy efficiency portfolio budget?

A. In the Companies' response to IR DPS-268, KEDLI proposes to allocate approximately $4.869 million of unspent EEPS funds to offset the proposed gas portfolio budgets in calendar years 2019 and 2020. Also in response to IR DPS-268, KEDNY proposes to allocate approximately $20.405 million of unspent EEPS funds to offset the proposed gas portfolio budgets in calendar years 2019 and 2020.

Q. Does the Panel recommend using any other source of funds to offset the gas energy efficiency budgets?

A. Yes. In addition to using unspent EEPS funds, we recommend using any interest that has accrued on the EEPS funds, unspent ETIP funds, and

interest accrued on the NYSERDA Clean Energy Fund (CEF) Bill-As-You-Go (BAYG) collections to offset the ETIP energy efficiency budgets.

Q. Please explain what is meant by “unspent ETIP funds.”

A. Unspent ETIP funds are those funds that the Companies collected from customers to pay for energy efficiency programs offered during the previous ETIP period but that the Companies have not spent as of the beginning of the Rate Year.

Q. How does interest accrue on NYSERDA CEF collections?

A. The NYSERDA CEF collections are collected by the Companies through the SBC surcharge. NYSERDA relies on a quarterly BAYG approach to obtain the CEF funding from the utilities. Under this methodology, KEDLI and KEDNY collect their authorized share of NYSERDA’s CEF budget on an ongoing basis, but only transfer their proportional share of NYSERDA’s actual expenditures to NYSERDA at the end of each quarter. This approach results in the Companies accruing a balance of CEF funds. Further, in accordance with the Commission’s Order

Authorizing the Clean Energy Fund Framework (CEF Order), issued on January 21, 2016 in Case 14-M-0094, the Companies are required to accrue carrying charges on this balance using the Other Customer Provided Capital Rate.

Q. What does the Panel recommend regarding the use of unspent EEPS and ETIP funds and EEPS and BAYG interest to offset the energy efficiency collections?

A. We recommend that the Companies use the combined total of the EEPS and ETIP unspent funds along with the EEPS and NYSERDA BAYG interest to offset the energy efficiency budgets over a five-year period. Exhibit__(SESP-2) provides a summary of the unspent EEPS and ETIP funds and EEPS and BAYG accrued interest and our recommended use of those funds.

Q. Does the change in the cost recovery method affect the manner in which the Companies implement the programs, the reporting requirements, or the implementation plan filing requirements currently associated with ETIP programs?

A. No. We recommend that the Companies continue to

implement the programs in accordance with the rules and requirements set forth in Commission orders and Staff guidance. Pursuant to the March 2018 ETIP Order the Companies are required to begin integrating energy efficiency planning into its forecasted system plans and evolve its ETIP into a SEEP, which will describe the entirety of the utilities expanded reliance on, and use of, cost effective energy efficiency to support its distribution system and customer needs. The method of cost recovery, whether through surcharges or base rates, should have no impact on the continued transparency of the utility-administered energy efficiency programs.

KEDLI HEAT Program

Q. Please provide the background of the KEDLI HEAT Program.

A. The Commission’s January 21, 2016 Order in the Clean Energy Fund proceeding, Case 14-M-0094, required that NYSERDA operate its programs in a fuel-neutral manner and, to facilitate this fuel-neutral approach, established a policy that eliminates collections from non-electric

customers. As KEDLI customers receive electric service from the Long Island Power Authority, which does not contribute to NYSERDA programs, KEDLI customers are not eligible for NYSERDA programs, including NYSERDA's EmPower Program for low-income customers. In order to avoid a gap in energy efficiency services to low-income customers in KEDLI's territory, the Commission required NYSERDA and KEDLI to pursue alternatives for 2017 and thereafter. KEDLI developed the HEAT program as its EmPower replacement program and has been implementing it since 2017.

Q. Please describe KEDLI's HEAT program.

A. The HEAT program is an energy efficiency program designed to serve low-income customers. The program consists of four tiers: Tier 1 includes customer outreach, education, and engagement to educate customers on the benefits of energy efficiency and encourage participation in Tier 2 measure offerings; Tier 2 consists of a home energy audit and direct installation of certain energy efficiency measures; Tier 3 entails the installation of additional energy efficiency

measures identified in the Tier 2 home energy audit; and Tier 4 includes the installation of custom energy efficiency measures that may require extensive work on the home, including the preparation of space needed to accommodate energy efficiency equipment.

Q. Does KEDLI propose any changes to the HEAT program?

A. Yes. On page 29 of the direct testimony of the Future of Heat Panel, KEDLI proposes to increase the annual budget from $1.9 million to $2.5 million. KEDLI also proposes to expand the program to include eligible moderate income customers whose residences do not have central air conditioning.

Q. Does the Panel support KEDLI's proposed expansions of the HEAT Program to moderate-income customers?

A. Yes, we support increasing the budget to $2.5 million and expanding the program to eligible moderate income customers.

Q. Is the HEAT program a part of KEDLI's energy efficiency portfolio?

A. No. Currently, KEDLI implements this program as

a standalone program that operates outside of the energy efficiency portfolio.

Q. Does the Panel recommend any changes to the way in which the HEAT program is implemented?

A. Yes. We recommend that KEDLI incorporate the HEAT program into its energy efficiency portfolio. At its core, the HEAT program is an energy efficiency program and KEDLI should operate this program in accordance with the rules and requirements associated with the current energy efficiency portfolio, including any statewide low-to-moderate income (LMI) customer framework developed within the Utility Energy Efficiency Proceeding. We further recommend the $2.5 million budget for the HEAT program be additional to the energy efficiency budget authorized by the Commission in Case 18-M-0084.

Q. Does KEDLI propose an incentive mechanism associated with the HEAT program?

A. Yes, KEDLI proposes the EmPower Replacement Program incentive mechanism, which would award KEDLI for customer participation in Tiers 3 and 4 of the HEAT program. The Staff EAM Panel

provides an overview and Staff's recommendations related to the KEDLI's proposed EmPower Replacement Incentive Mechanism.

Energy Efficiency Labor

Q. Are the Companies proposing any incremental full time equivalent employees (FTEs) associated with the energy efficiency programs?

A. Yes. On page 28 of the direct testimony of the Future of Heat Panel, the Companies propose an additional seven FTEs: four FTEs focused on product development and delivery to customers, one FTE to support sales activity for commercial and industrial products and two FTEs for technical sales support.

Q. Does the Panel support KEDNY and KEDLI's request for additional FTEs?

A. Yes. The Companies are significantly ramping up energy efficiency-related initiatives. Specifically, the KEDNY energy efficiency target increases by 32% in 2019 and 55% in 2020 and for KEDLI the energy efficiency target increases by 26% in 2019 and 45% in 2020. Therefore, we agree that it is reasonable for the Companies to

add seven additional FTEs to help support the enhanced energy efficiency portfolios.

UTILITY ENERGY SERVICE CONTRACT PROGRAM

Q. Please describe the Utility Energy Service Contract (UESC) Program.

A. As described on pages 34 and 35 of the direct testimony of the Future of Heat Panel, the Companies have operated the UESC Program since 2011 and it is an energy services contracting program whereby the Companies and customers enter limited-source contracts for energy management services. The customer pays for the improvements delivered under the UESC Program. The Companies have been offering the UESC Program through the Federal Energy Management Program that targets government facilities. The UESC Program has since evolved to include large commercial customers as well.

Q. Are the Companies proposing changes to the UESC Program?

A. Yes. As described on pages 35 and 36 of the direct testimony of the Future of Heat Panel, the Companies propose changes including: (1)

incentivizing up to 50 percent of the total cost of a feasibility study that assesses energy savings opportunities, calculates the costs of those savings recommendations, and provides implementation assistance at a customer's facility; (2) adopting a more proactive approach to attract participants; (3) increased involvement in the development of project scopes for participating customers; and (4) implementing a Platform Service Revenue fee structure.

Q. Does the Panel support the proposed changes?

A. Overall, we support the changes. However, this Panel will not address the most significant change of the program, the Platform Service Revenue fee structure. The Platform Service Revenue fee structure is discussed in the testimony of the Staff EAM Panel.

Q. Does the Panel have any additional recommendations regarding the UESC program?

A. Yes, we recommend that the Companies be required to develop specific energy efficiency targets and budgets for the UESC Program as part of the ETIP/SEEP and include the program as a part of

the Companies' overall energy efficiency portfolio.

Q. Please explain why the Panel recommends developing energy efficiency targets and budgets for the UESC Program?

A. The UESC program incorporates energy efficiency measures and results in energy savings and should therefore be transitioned into the Companies' energy efficiency portfolios to ensure consistent and comprehensive treatment of energy efficiency programs and their resultant savings. We recommend that the UESC program adhere to all implementation and reporting requirements associated with the Companies' ETIP/SEEP Programs.

Q. How does the Panel recommend the Companies fund the program?

A. We recommend that the Companies fund the UESC Program with the authorized energy efficiency budgets. Historically, the Companies have not spent the annual energy efficiency budgets and can accommodate the UESC program budget without additional funding beyond that of the energy efficiency portfolio budgets.

Q. Are the Companies proposing additional FTEs for the UESC program?

A. Yes. The Companies proposed two additional FTEs: a marketing manager and a project engineer.

Q. Does the Panel agree with the additional FTEs?

A. No. We do not support additional FTEs for the UESC program as the Companies have not sufficiently justified the need for the additional FTEs they are requesting. As we recommend that the Companies shift the UESC program into the energy efficiency portfolio, we recommend that the Companies rely on the existing and incremental FTEs associated with the energy efficiency portfolio to implement the UESC program.

FUEL SWITCHING CALCULATOR

Q. Please describe the fuel-switching calculator as proposed by KEDNY and KEDLI.

A. On page 38 of the direct testimony of the Future of Heat Panel, the Companies describe a proposed fuel-switching calculator as a web-based tool that would inform customers about the energy

savings and financial implications associated with switching from the customer's existing fuel source to an alternative fuel source, including natural gas, geothermal heat pumps, and renewable natural gas. The calculator would provide customers with a side-by-side comparison, including the estimated annual cost, payback period, available incentives, tax credits, conversion costs, and annual fuel savings.

Q. Does the Panel support the fuel switching calculator?

A. Yes. The fuel-switching calculator enables customers to evaluate different heating options to make more informed energy decisions when choosing or replacing the energy equipment in their homes.

Q. Does the Panel have any recommended changes associated with the fuel-switching calculator?

A. Yes, we have two recommendations. First, we recommend that the Companies' adopt standardized assumptions and calculations where appropriate. Second, we recommend that the calculator include other end-uses of fuel in addition to primary

heat such as domestic hot water, cooking, or air conditioning in order to give customers a complete picture of additional energy efficiency upgrades that can be made.

Q. Does the Panel have any other concerns with the fuel-switching calculator?

A. Yes. The Fuel-Switching Calculator is a tool that could be used more broadly by National Grid's operating utilities to inform their customers and, therefore, the $250,000 cost of the Fuel-Switching Calculator should be shared among other National Grid entities. This would reduce costs to KEDNY and KEDLI. To appropriately share the cost with other National Grid entities, we recommend using the general allocator to assign 8.29 percent of the cost to KEDLI and 13.31 percent of the cost to KEDNY.

DEMAND RESPONSE

Q. Please describe the Companies' Demand Response (DR) programs.

A. On page 42 of the direct testimony of the Future of Heat Panel, the Companies state that they have two DR programs. The first is the

Interruptible and Temperature Control (IT/TC) Service that offers a reduced delivery rate for customers that interrupt their natural gas service when demand peaks. The IT/TC service has two pricing tiers that customers are placed into based on the customer's fuel switching capabilities. The Companies do not propose changes to IT/TC service in this proceeding.

Q. What is the second DR program?

A. The second DR program is currently a Demonstration Project providing market-based payment to commercial firm customers who commit to minimum gas usage reductions during peak events, based on their performance during specified DR events. This project was approved as part of the 2016 Rate Order, issued on December 16, 2016, in which the Commission adopted a Joint Proposal and established rate plans for the Companies in Cases 16-G-0058 and 16-G-0059. Our testimony in these proceedings will focus on the Companies' proposals related to this program, and we will refer to this program as the Performance-based Gas DR Program.

Q. How do customers currently participate in the

demonstration-scale program?

A. Currently, customers enroll load relief associated with direct utility control of specific equipment within customers' premises, with an associated bid price. The Companies use a competitive selection process based on the load relief and price proposed in the bids received from customers. The Companies develop a load relief portfolio seeking to maximize load relief and minimize costs. Thereafter, during a load relief event, KEDNY and KEDLI directly shut off enrolled customer equipment for the duration of the Event. A more complete description of the current Gas Demand Response Demonstration Project is available as part of the Companies' implementation plan filings made on July 27, 2017 in Cases 16-G-0058 and 16-G-0059.

Q. Do KEDNY and KEDLI propose changes to the Performance-based Gas DR Program?

A. Yes. On pages 44 and 45 of the direct testimony of the Future of Heat Panel, the Companies propose scaling the project up from demonstration size and including the costs of the Performance-based Gas DR Program in base

rates. Additionally, the Companies propose making three operational changes which will impact participant responsiveness to peak events.

Q. What is the first operational change that the Companies propose?

A. The Companies propose certain new requirements for customer sizes and proposed load relief amounts. Specifically, the Companies to require that firm customers meet a minimum annual demand value, that participants provide a minimum reduction value to be eligible to participate in the program, and that the amount of load relief pledged by participants must be at least 10 percent of either the customer’s gas usage during a peak period or the sum of such customer’s nameplate gas usage requirements. Specifically, KEDNY proposes that firm customers must consume at least 6,000 Dth per year, and must pledge a minimum load relief amount of 2 Dth/hour during events. Similarly, KEDLI proposes that firm customers must consume at least 4,000 Dth per year, and must pledge a minimum load relief amount of 2 Dth/hour during

events.

Q. What is the second operational change that the Companies propose?

A. The Companies propose to require that, in order to receive an incentive payment, participants must reduce their gas demand for the entire duration of the 3-hour peak event for each event called.

Q. What is the third operational change that the Companies propose?

A. The Companies propose to impose a penalty when participants do not provide their enrolled load relief, and may seek or revoke customers' eligibility to participate in the program if the participating customers fail to respond to peak events.

Q. Does the Panel support these changes?

A. Yes. These changes would more closely align the Companies' Performance-based Gas DR Program with other Electric DR programs focused on reducing Commercial and Industrial customers' peak demand, and would closely align with a similar performance-based Gas Demand Response pilot currently being operated by Con Edison.

Aligning KEDNY and KEDLI's program rules with other programs targeting the same customer segment should help the Companies attract customers that may already be familiar with the Electric demand response programs, and allow participants with greater flexibility regarding the actions they take to reduce gas demand when called upon since customers will be free to choose how to provide load relief instead of enrolling specific pieces of equipment for utility control. Overall, we generally agree that these changes will likely result in a more successful DR program.

Q. Does the Commission typically require a significant degree of identical requirements for DR programs across utility service territories?

Q. Yes, however, although there are a number of minor differences between the Performance-based Gas DR Program proposed by the Companies and Con Edison's Gas DR Pilot, these differences are reasonable at this time.

Q. Why?

A. Unlike the Electric DR programs, which have been operational across the major electric investor-

owned utilities since the summer of 2015, and since 2009 at Con Edison, gas DR programs designed for firm service customers are in their infancy. The program proposed by KEDLI and KEDNY will be only the second such performance-based gas demand response program in New York State. Therefore, although the Commission has required a significant level of consistency among the Electric DR programs, it makes sense to allow KEDNY and KEDLI to test the effectiveness of several program features against Con Edison's similar program.

Q. What are the capital expenditure (CapEx) costs related to the Performance-based Gas DR Program?

A. As stated by the Companies on pages 47 and 48 of the Future of Heat testimony, for KEDNY, the CapEx costs of the Performance-based Gas Demand Response Program are $236,000 during the 12-month period ending March 31, 2020, with additional CapEx costs of $59,000 per year during the Rate Year, and Data Years 1 and 2. For KEDLI, the CapEx costs are $107,000 during the 12-month period ending March 31, 2020, with additional CapEx costs of $27,000 per year

during the Rate Year, and Data Years 1 and 2.

Q. What are the Operations and Maintenance (O&M) costs related to the Performance-based Gas Demand Response Program?

A. O&M costs related to the program include both labor and non-labor O&M. For non-labor O&M costs at KEDNY, the Companies identify $1.216 million during the Rate Year, and $1,531 million, $1,743 million, and $2.104 million during Data Years 1 through 3, respectively. For non-labor O&M costs at KEDLI, the Companies identify $0.577 million during the Rate Year, and $0.720 million, $0.814 million, and $1.004 million during Data Years 1 through 3, respectively. These forecast non-labor O&M costs would be recovered through base rates. For labor costs, the Companies propose to hire one incremental FTE during both the Rate Year and Data Year 1, with an additional two FTEs each year for Data Year 2 and Data Year 3. The Companies propose to split these FTEs 70/30 between KEDNY and KEDLI.

Q. Does the Panel agree with the Companies' cost forecasts and cost recovery mechanisms?

A. We do not support the Companies' proposal to hire additional FTEs for this program, and we recommend an alternate cost recovery mechanism for the non-labor O&M costs.

Q. Why does the Panel not agree with the Companies' request for additional FTEs?

A. The Companies already have personnel assigned to the Demand Response Demonstration Project, which should be sufficient to run the Performance-based Gas Demand Response program given the simultaneous and significant investments the Companies propose to make for management software and automated systems to help manage this program. Therefore, we recommend eliminating the Companies' requested additional FTEs for this program.

Q. Why does the Panel disagree with the Companies' proposal to include non-labor O&M costs of the Program-based Gas Demand Response program in base rates?

A. Non-labor O&M costs related to most demand response programs are related to incentive costs paid to participants. As these incentive payment costs can vary significantly since they

are based on the number of participants enrolled in the program, the amount of load relief enrolled in the program, the number of events called in a given year, and, of course, participant performance during those events. These factors can lead to significant swings in incentive payment costs from year to year, and can be difficult to forecast. Therefore, we recommend that, instead of recovering these costs in base rates, the Companies recover the non-labor O&M costs as-incurred through the Delivery Service Adjustment surcharge.

Q. Do other utilities recover the costs of their demand response programs in a similar way?

A. Yes. The other investor-owned utilities recover the costs of their respective demand response programs through a non-bypassable delivery surcharge mechanism. Applying this same cost recovery mechanism to KEDNY and KEDLI would bring their programs in line with other electric and gas demand response programs statewide.

SUSTAINABLE HEAT INITIATIVES

Expanded Geothermal Demonstration Pilot Program

Q. Please summarize KEDLI's Geothermal Demonstration Project, which was implemented through the 2016 Rate Order.

A. The KEDLI Geothermal Demonstration Project, described on pages 50-52 of the direct testimony of the Future of Heat Panel, was established to test a ground-source heat pump solution using a shared-loop system and its ability to provide heating and cooling to homes in Glenwood Village, an area not currently served by KEDLI's gas network. This test-and-learn demonstration project provides ten homes with an alternate heating and cooling and energy cost savings through a geothermal loop system that is owned by the Company. The geothermal system is operational in the Glenwood Village community and KEDLI is moving forward with the demonstration project at a site in Medford, where construction will begin in the third quarter of 2019.

Q. Please describe the performance of the Geothermal Demonstration Project to date.

A. KEDLI'S quarterly report for the Geothermal Demonstration Project, covering the period from April 1, 2019 through June 30, 2019, filed July 31, 2019 in Case 16-G-0058, states that the Glenwood Village geothermal system continues to be monitored to ensure that it is functioning properly; and that performance data continues to be collected and analyzed. The Medford geothermal system is at the beginning stages of installation.

Q. Does KEDLI propose changes to the Geothermal Demonstration Project?

A. Yes. On page 52 of the direct testimony of the Future of Heat Panel, the Companies propose to expand the project to other parts of KEDLI's service territory and to KEDNY's as well, prioritizing LMI and commercial and industrial customers. The Companies propose to charge a fixed monthly fee to participating customers for the design and the installation of the geothermal system that will be owned by KEDLI or KEDNY. The Companies state that utility-ownership of the ground loop will increase the adoption rate of this type of technology.

Q. Does the Panel support the expanded Geothermal Demonstration Project?

A. No, not at this time. The existing Geothermal Demonstration Project is just entering the data analysis phase and has not yet been proven to be cost-effective. In addition, KEDNY and KEDLI have not demonstrated that utility ownership of a geothermal system would not adversely affect third-party market participants that may be able to provide additional affordable alternatives. Finally, the Companies have not performed a solicitation, such as for a non-pipeline alternative, seeking third-party developers to demonstrate that there is a need for utility ownership.

Q. Do the Companies request any incremental FTEs associated with this program?

A. Yes, the Companies have requested one FTE to support the expanded geothermal program.

Q. Does the Panel support the proposed incremental FTE?

A. No. We do not support the Companies' proposal for an additional FTE as we do not support expanding the Geothermal Demonstration Project.

Clean Conversion Program

Q. What is the Clean Conversion Program?

A. KEDLI proposes to replace the existing Neighborhood Expansion Program with the Clean Conversion Program. Under the Clean Conversion Program, KEDLI proposes to work with customers that request a new gas connection to educate them about their environmentally friendly energy options. If the customer does not choose an environmentally friendly non-gas option, KEDLI will offer an opportunity to switch from oil to natural gas. Under the Clean Conversion Program, KEDLI will extend gas service to customers under the same density and minimum customer connections requirements established in the Order Directing Investments and Allowing, In Part, Deferral Authority for Costs Associated with Incremental Capital Expenditures and Establishing a Surcharge, issued on December 15, 2014 in Case 14-G-0214.

Q. What is the proposed cost of the Clean Conversion Program?

A. KEDLI proposes $26 million in CapEx in the Rate Year, $23 million in Data Year 1, $19 million in

Data Year 2, and $16 million in Data Year 3, as shown in Exhibit___(GIOP-1), Schedule 1.

Q What are the differences between the Neighborhood Expansion Program and the Clean Conversion Program?

A. As detailed in the response to IR DPS-596, the Clean Conversion Program is primarily a reactive program that builds off of the Neighborhood Expansion Program's customer interest, density, and minimum threshold customer participation requirements that is applied when a customer requires natural gas facilities, whereas the Neighborhood Expansion Program is meant to take a proactive approach into identifying areas of potential large project expansion.

Q. Does the Panel have any issues with the KEDLI's proposal?

A. Yes.

Q. Please explain.

A. As explained by the Staff Policy Panel, Staff's position in these rate proceedings reflects an assumption that the Northeast Energy Supply Enhancement (NESE) project will not be built during the Rate Year. Without the NESE Project,

KEDLI’s ability to connect new customers is limited. Consequently, a program based upon potential new natural gas customers will not be of significant use. We recommend not approving the Clean Conversion Program or its budget.

LMI Conversion Program

Q. Please describe the Companies LMI Conversion Program

A. On page 59 of the direct testimony of the Future of Heat Panel, the Companies propose to expand the low-income conversion program to moderate income customers. The program pairs energy conversions with incentives for high efficiency heating and hot water equipment to reduce a customer’s carbon footprint and energy bills.

Q. Would the Companies pursue the LMI Conversion Program if the NESE project does not get built?

A. In its supplemental filing on June 11, 2019 in which the Companies discussed the implications of NESE not being built on the proposals in these rate proceedings, the Companies indicated that they would not implement the LMI Conversion Program.

Q. What is the Panel's position regarding the LMI Conversion Program?

A. Given that, for the purposes of these rate proceedings, Staff is assuming that NESE will not be built during the Rate Year, we accept the Companies' position to not implement the LMI Conversion Program.

REDUCING CARBON EMISSIONS ON THE GAS NETWORK

Renewable Natural Gas Interconnections

Q. Please describe what KEDNY and KEDLI propose regarding renewable natural gas (RNG) interconnections.

A. As described on page 89 of the Testimony of the Future of Heat Panel and set forth in KEDNY and KEDLI's GIOP testimonies, the Companies propose to facilitate RNG interconnections by installing and owning the interconnection facilities of successfully implemented RNG developer projects and also propose to create a new Future of Heat Engineering department that will include additional FTEs to review and process future interconnection applications.

Q. What does the Panel recommend regarding the RNG

interconnection proposal?

A. While we support this proposed program, KEDNY's Rate Year RNG projects are still in the infancy stages of development. Additionally, there is the potential reallocation of FTE's due to our recommended capital program changes associated with the Clean Conversion Program. Accordingly, we recommend denying the additional requested FTEs for both Companies and remove the associated O&M related to the KEDNY projects for the Rate Year. The associated KEDNY Rate Year CapEx should also not be allowed and will be addressed by the Staff Gas Infrastructure and Operations Panel.

Power-to-Gas Demonstration Project

Q. Please describe what KEDLI proposes for the power-to-gas demonstration project.

A. KEDLI, as described on page 83 of the Testimony of the Future of Heat Panel, proposes to develop a project that converts excess renewable energy to hydrogen and then converts the hydrogen to pipeline-quality natural gas to be ultimately injected into the existing gas network and

delivered to customers. The Company proposes to begin scoping, design, and engineering of the project in 2019 with an anticipated full project design completion in October of 2023. The projected cost of the project scoping and design phase includes O&M costs of $0.65 million in the Rate Year, $1.825 million in Data Year 1, $0.725 million in Data Year 2, and $0.325 million in Data Year 3. KEDLI would propose project construction costs and costs in excess of those above in its next rate case.

Q. Does the Panel support this project?

A. No. The current energy storage targets set forth by the Commission in its Order Establishing Energy Storage Goal and Deployment Policy (Storage Order), issued on December 12, 2018 in Case 18-E-0130, provide for the envisioned solutions and goals for excess renewable electricity. We recommend that the Commission deny the project and the associated funding.

Hydrogen Blending Study

Q. Please describe what the Companies propose for the hydrogen blending study.

A. As described starting on page 87 of the Testimony of the Future of Heat Panel, the Companies propose a hydrogen blending study to analyze the impact of blending hydrogen into the natural gas system, to determine the percentage of hydrogen that can be safely blended into the Companies' gas network without negatively impacting the network or customer appliances. This would be a precursor to the development of a hydrogen blending demonstration project. The projected cost of the study would be $0.446 million in the Rate Year split evenly between KEDNY and KEDLI, assuming no additional grant funds are received.

Q. Does Staff support this project?

A. No. As with the Power-to-Gas Project, the current energy storage targets set forth in the Storage Order provide for the envisioned solutions and goals for excess renewable electricity. Therefore, there is no need at this time to study utilizing that excess

renewable electricity to inject hydrogen into the Companies' gas networks. Accordingly, we recommend that the Commission deny this project and the associated funding.

Q. Does this conclude your testimony at this time?

A. Yes.